

Evidence Insufficiency in Relativistic Physics Queries: A Procedural Meta-Analysis of Allowed Corpus EV-Q074

Input Question

Will we ever travel at the speed of light?

Domain

Physics

Validation Status

needs_data

Problem Statement

The question asks whether humans or technology can achieve travel at the speed of light, constrained by current physical understanding, technological needs, and biological limitations. However, the provided evidence corpus contains no information regarding relativistic physics, mass-energy equivalence, or cosmic speed limits.

Rationale

A rigorous scientific answer requires evidence grounded in Special Relativity or related physical theories. The allowed evidence IDs (EV-Q074-a9c4f41349a15827a20ae191, EV-Q074-35e2708bd2ab08c48151e39f, EV-Q074-08330f45723912c144cbc114, EV-Q074-28b8cac19d8d0787c4792449) exclusively discuss network latency, carbon footprint accounting, robotic exploration metrics, and LLM-based travel planning. Therefore, the only scientifically valid conclusion based on the allowed evidence is that the corpus is insufficient to answer the physics question. This report formulates a procedural meta-hypothesis to validate this insufficiency.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: Procedural Meta-Hypothesis: The allowed evidence corpus (EV-Q074-a9c4f41349a15827a20ae191, EV-Q074-35e2708bd2ab08c48151e39f, EV-Q074-08330f45723912c144cbc114, EV-Q074-28b8cac19d8d0787c4792449) contains zero semantic content relevant to relativistic physics or light-speed travel constraints, rendering scientific hypothesis generation on this topic impossible within current bounds.
- **Mechanism**: This is a procedural finding of evidence insufficiency, not a physical theory. The mechanism is the strict adherence to evidence-grounding protocols which prohibit synthesizing physics claims from unrelated domains (network latency, carbon accounting, robotics, LLM planning). The 'result' is a validated knowledge gap rather than a causal explanation of light-speed barriers.

- ****Falsifiable Prediction****: If a validated semantic search protocol (including a positive control using known physics literature) detects any mention of 'Special Relativity', 'mass-energy equivalence', 'Lorentz factor', or 'vacuum speed limit' in the full text of the four allowed evidence IDs, this meta-hypothesis is falsified.
- ****Required Observations****: Full-text semantic verification of all four allowed evidence IDs confirming absence of relativistic physics terminology ; Positive control test: The same search protocol must successfully identify relativistic physics terms in a known physics paper (e.g., Einstein 1905 or standard textbook excerpt) to prove method sensitivity ; Domain classification confirming each allowed ID belongs strictly to Network/Environmental/Robotics/AI domains
- ****Risk of Being Wrong****: Moderate risk if the semantic search protocol fails due to poor keyword selection or embedding model limitations; mitigated by mandatory positive control. Low risk of factual error regarding physics since no physics claims are made.

Hypothesis 2

- ****Hypothesis****: Semantic Polysemy Hypothesis: The term 'travel' in the allowed evidence IDs exclusively refers to non-relativistic domains (network packet latency, aviation carbon footprint, robotic path length, or LLM itinerary planning), and never to relativistic mass transport.
- ****Mechanism****: Lexical ambiguity resolution where 'travel' is a polysemous term. The mechanism posits that without explicit physics context markers (e.g., 'c', 'gamma', 'inertial frame'), the default semantic cluster for 'travel' in these documents is computational or logistical, not physical. This hypothesis serves as a negative validation step for the meta-hypothesis.
- ****Falsifiable Prediction****: If any sentence in the allowed evidence IDs uses 'travel' in conjunction with physics-specific modifiers (e.g., 'near-light travel', 'relativistic travel', 'photon travel'), this polysemy hypothesis is falsified.
- ****Required Observations****: Contextual embedding analysis of every instance of 'travel' in the four allowed IDs ; Clustering visualization showing 'travel' instances grouping with network/robotics/AI terms, not physics terms ; Manual annotation of ambiguous cases to confirm non-physics usage
- ****Risk of Being Wrong****: Low risk, as the quoted texts in evidence cards already strongly suggest non-physics contexts. Primary risk is metaphorical usage (e.g., 'light' as weight) being misclassified, but this does not constitute physics relevance.

Technical Details

This experiment validates the Procedural Meta-Hypothesis that the allowed evidence corpus contains zero semantic content relevant to relativistic physics. The protocol involves: 1) Full-text extraction from the four allowed Evidence IDs (EV-Q074-a9c4f41349a15827a20ae191, EV-Q074-35e2708bd2ab08c48151e39f, EV-Q074-08330f45723912c144cbc114, EV-Q074-28b8cac19d8d0787c4792449). 2) Semantic search for a predefined lexicon of relativistic physics terms (e.g., 'Special Relativity', 'Lorentz factor', 'mass-energy equivalence', 'vacuum speed limit'). 3) Contextual classification of the term 'travel' in each document to confirm it refers to network latency, carbon accounting, robotic exploration, or LLM planning. 4) Positive Control Test: Application of the same search protocol to a known physics text (e.g., Einstein's 1905 paper excerpt) to verify the method's sensitivity and ability to detect relevant content if present. 5) Negative Result Verification: Confirmation that no physics terms are found in the allowed IDs, thereby validating the knowledge gap.

Datasets

Source

```
```json
[
{
 "id": "EV-Q074-a9c4f41349a15827a20ae191",
 "description": "arXiv:2002.10263 - Network travel time diversity."
},
{
 "id": "EV-Q074-35e2708bd2ab08c48151e39f",
 "description": "arXiv:2004.05603 - Carbon footprint of flight travels."
},
{
 "id": "EV-Q074-08330f45723912c144cbc114",
 "description": "arXiv:2404.19564 - Robot exploration metrics."
},
{
 "id": "EV-Q074-28b8cac19d8d0787c4792449",
 "description": "arXiv:2607.26977 - LLM-based travel planning."
}
]
```
```

Target

Semantic verification report confirming absence of relativistic physics content; Positive control validation log.

Paper Abstract

Background: The question of light-speed travel is fundamentally constrained by Special Relativity. However, automated evidence retrieval systems may return semantically irrelevant documents sharing lexical overlap (e.g., 'travel'). Methods: We formulate a procedural meta-hypothesis that the allowed evidence corpus (four arXiv papers on network latency, carbon footprints, robotics, and AI planning) contains no relevant physics content. We design a validation experiment involving full-text semantic search for relativistic terms and a positive control

test using standard physics literature. Verification Plan: Execute NLP pipeline to confirm zero hits for physics terms in allowed IDs and high recall in positive control. Results: pending (待执行验证实验).

Methods

1. Text Extraction: Use PyPDF2 to extract full text from the provided URLs of the four evidence IDs and the positive control physics text. 2. Keyword Matching: Perform exact and fuzzy matching for a curated list of 50+ relativistic physics terms. 3. Semantic Embedding: Use a pre-trained transformer model (e.g., BERT) to generate embeddings for sentences containing ambiguous terms like 'travel'. 4. Contextual Classification: Cluster embeddings to distinguish between physics contexts and non-physics contexts. 5. Positive Control Validation: Run the same pipeline on a known physics paper excerpt to ensure recall > 0.95 for physics terms.

Experiments

Baselines

```
```json
[
 "Random Keyword Search: Simple string matching for 'speed of light' without context awareness.",
 "General Domain Classifier: A naive Bayes classifier trained on general news vs. physics papers to estimate domain probability."
]
```
```

Metrics

```
```json
[
 "Precision of Physics Term Detection: Ratio of true physics-related mentions to total mentions of ambiguous terms.",
 "Recall on Positive Control: Percentage of known physics terms correctly identified in the control text.",
 "Domain Classification Accuracy: Accuracy of classifying each allowed ID into its actual non-physics domain."
]
```
```

Ablation

Remove contextual embedding analysis and rely solely on keyword presence to demonstrate the risk of false positives (e.g., 'light' in 'lightweight robots').

Validation Protocol

Double-blind manual review of 10% of the extracted sentences by two independent human annotators to verify the automated classification of 'travel' context and confirm the absence of physics relevance.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- ****EV-Q074-a9c4f41349a15827a20ae191**** · arxiv · arXiv:2002.10263
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2002.10263.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:7|section:page-7|paragraph:1;
content_sha256=0472c4df431dca0a0323097c3c986b5a8299082ce14808ddd8cc6deb1d5569e6
- ****EV-Q074-35e2708bd2ab08c48151e39f**** · arxiv · arXiv:2004.05603
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2004.05603.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=8db4992f0e8eb7d1cc1b2d9fdb3e0d76e239174f01ce5256145aece7bd0111cd
- ****EV-Q074-08330f45723912c144cbc114**** · arxiv · arXiv:2404.19564
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2404.19564.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=f06034c6b6f6cca90a7030cceb77380de1551b97d22423a6524a3c50c3786acd
- ****EV-Q074-28b8cac19d8d0787c4792449**** · arxiv · arXiv:2607.26977
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2607.26977.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=1d85b917cc31f9c560eaea328038e82096c0b4b5276dbd17bde47c59fed2ed5a

Reviewer Comments

- The candidate hypothesis correctly frames the output as a 'Procedural Meta-Hypothesis' regarding evidence insufficiency, explicitly avoiding any claim of being a physical theory about light-speed travel. This satisfies required_revision:7761f20ad5c4.
- The experiment design now includes a mandatory positive control (known physics paper) to validate the sensitivity of the semantic search protocol, satisfying required_revision:4dbf5e906c38.
- All supporting_evidence_ids are valid and present in the allowed EvidenceCatalog; no external or fabricated IDs are used.
- The Results field correctly states 'pending' and does not fabricate experimental outcomes.
- The falsifiable prediction is logically sound: finding relativistic physics terms in the allowed corpus would indeed falsify the meta-hypothesis of irrelevance.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Verify access to all four arXiv PDF URLs listed in EvidenceCatalog.
- Ensure the keyword list for relativistic physics is comprehensive and documented.
- Include a standard physics text excerpt as a positive control for method validation.
- Save the raw text extraction outputs for audit.
- Document the version of the embedding model used for semantic analysis.
- Store the manual annotation guidelines for the validation protocol.